

Mobile Operator Application

Predictive Analytics Service

Pre-drying
prediction (ML)

Dynamic remaining
time (ML)

Spoilage-risk
rule

Over-drying
auto-stop

Safety clamp + orchestration

Batch & Sustainability Service

Batch record
(fish, weights, salt, status)

Waste / salt
estimation

Traceability
reports

Recycler notification

HTTP API

stop (safety)

Environmental Monitoring Service (edge)

Drying state
machine

AUTO control
(hysteresis)

Sensor validate
-> FAULT

Alerts +
event log
Auth
Service

Serial link | actuator protocol | mock-sensor mode

Drying Chamber (edge hardware)

temp/humidity · temp probe · gas · load cell · relays

Heater / Exhaust
fan / Light